From Reliability to Resilience: An Adaptive Operational Margin Framework for UAV C2 Communication

Shreeja Sridharan
Deutsche Telekom Chair of Communication Networks
Technische Universität Dresden
01217 Germany

Enrique Caballero
Institute of Networked and Embedded Systems
University of Klagenfurt
9020 Austria

Stanislav Mudriievskyi
Deutsche Telekom Chair of Communication Networks
Technische Universität Dresden
01217 Germany

Christian Bettstetter
Institute of Networked and Embedded Systems
University of Klagenfurt
9020 Austria

Frank H.P. Fitzek
Deutsche Telekom Chair of Communication Networks
Centre for Tactile Internet with Human-in-the-Loop (CeTI)
Technische Universität Dresden
01217 Germany

***Abstract*—**text from abstract As the low-altitude economy accelerates, integrating UAVs into non-segregated airspace demands more than 'best-effort' wireless connectivity. Moreover, reliability and resilience within the digital communication domain is now an urgent priority due to the unmanned and autonomous nature of UAVs. The Command and Control (C2) communication system, an essential safety tether for Beyond Visual Line of Sight (BVLOS) operations requires a rigorous framework to meet strict aviation specifications, deterministically reliable and resilient. The existing Required Communication Performance (RCP) defines reliability by setting strict static bounds on parameters like Transaction Time, Continuity, Availability, and Integrity. Beyond the scope of security resilience, C2 system requires a dedicated focus on communication and connectivity: *verifying whether the C2 link is active or impending degradation in C2 message traffic*. Therefore, in this work, we address the gap from C2 reliability to system-level resilience by focusing only on communication and connectivity paradigm. To achieve this, we introduce a **behavioral connectivity resilience model**, resilience in connectivity, calibrating system behavior to the operationally functional boundary is built upon two key contributions. First, we develop a phase-based C2 system architecture spanning proactive, reactive, and post-corrective operational windows to dynamically manage the RCP under potential C2 degradation condition. The Proactive Phase anticipates and preemptively adjust transmission mechanisms on the C2 system. The Reactive Phase actively mitigates real-time network delays to maintain vehicle control during active link stress, while the Post-Corrective Phase safely governs system resynchronization and state restoration. Second, we formulate a dynamic Operational Margin representation that envelopes these three phases specifically for Very Low Level (VLL) airspace. This time-variant behavioral framework is evaluated using a UAV equipped with a heterogeneous, multi-link communication suite (WiFi, 5G, and LEO Satcom) compliant with Specific Assurance and Integrity Level III (SAIL III) and above. By analyzing performance metrics across each operational window, we present the derived operational margins for the evaluated system, demonstrating that the UAV can safely navigate communication anomalies through adaptive, behavioral control of the C2 system.

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1. Introduction

UAVs operating beyond visual line of sight (BVLOS) rely on the Command and Control (C2) link. When it degrades, they enter in a Lost C2 Link (LC2L) state [1].

Citation plan: [2], [3]

2. Related Work

RCP and its limits

ICAO's PBCS framework [4] introduced the performance-based paradigm for aviation communication. Any link Required Communication Performance (RCP) formalism defines communication requirements based on four parameters: transaction expiration time, continuity, availability, and integrity. RCP itself is not specified for any technology, is technology-agnostic. It constrains the transaction outcome, not the delivery mechanism. Gómez Depoorter and Lücke refined this for data-link design. They isolated the technical component from the human one to create Required Communication Technical Performance (RCTP) [5]. They later applied RCTP to air-ground data link design [6] and to model and evaluate RCP compliance [7]. Klügel *et al.* showed how to measure and model ICAO's PBCS metrics on a real system [8]. Earlier, they derived communication demands and performance metrics for next-generation aerial networks [9].

For unmanned aviation, aeronautical authorities have developed concepts through the last ten years. JARUS produced two documents: the RPAS C2 Link RCP concept [10], which frames C2 link performance in RCP vocabulary, and the later Required C2 Link Performance (RLP) concept [11], which SORA subsequently adopts as the parameter set proper to the C2 link, reserving RCP for communication with air traffic control.

For the present work, we extract two main properties in this study. First, every parameter is a *bound*: a transaction either completes before its deadline or not, and continuity measures the success probability over a population of transactions. Second, these bounds are *static*. The framework budgets a nominal transaction ignoring how the system behaves when it exceeds this budget or how long recovery takes. RCP therefore specifies steady-state adequacy, not behaviour under disturbance. This limitation remains hidden while the link works or fails. In this work, we treat degradation as a time-varying condition, thus the dynamics become critical.

Aerial C2 links

Early 3GPP studies on aerial LTE [12] identified key aerial channel characteristics: line-of-sight dominance at altitude, uplink interference and unstable cell association. Extensive campaigns quantified these effects. Amorim *et al.* measured the reliability for aerial vehicles [13], [14]. Nguyen *et al.* explored how to maintain this connectivity [15]. Other campaigns built latency profiles and models for cellular UAV control [16], [17], verified by latency measurements in operational networks against 3GPP [18], [19]. Recently, researchers have also surveyed non-terrestrial and hybrid architectures [20], [21], characterizing LEO latency and handovers between terrestrial and non-terrestrial segments [17], [22].

This research sets the baseline for evaluation. For the present work, it reports distributions: latency percentiles, outage rates, coverage maps. A latency CDF cannot explain why one system degrades gradually and recovers quickly from one that fails abruptly and recovers slowly. However, these two scnearios are two different problems for a remote pilot.

Resilience formalism

Outside aviation, researchers quantify resilience as a time-dependent system. In networking, Sterbenz *et al.* formulate the defend, detect, remediate, recover; diagnose, refine strategy (D^2R^2+DR) and its evaluation methodology [23]. At a network level [24], resilience is evaluated, but not at the link level. In power systems, building on earlier modeling [25] and expanding foundational work [26], Panteli *et al.* introduce the resilience trapezoid, modeling explicit degradation, degraded-state and recovery phases, providing metrics for each [27]. IEEE PES task force later consolidated the methodology [28].

Reactive— Mechanisms to sustain a link under stress are mature. Key approaches include multi-connectivity [29], packet duplication [30], and handover outage mitigation [31]. Three contributions closely relate to our work. Colpaert *et al.* demonstrate multi-operator connectivity for drone delivery [32]. Salehi *et al.* analyse reliability and delay across a 3D satellite/HAPS/cellular network using multi-connectivity and QoS-aware path selection [33]. Topologically, they use the same heterogeneous suite we consider here. Sridharan *et al.* propose a reliable multi-link framework for RPAS C2 communication and quantify 5G and LEO performance [34]. Other studies are on multipath transport over cellular and LEO [35], latency-bounded multipath protocols [36], and erasure coding for ATC communication [37], [38].

Table 1. Gaps for our work (To-be-deleted)

#	Gap	What should we do
G1	No criteria for when a re-established C2 link may resume normal control	Unique selling point for us
G2	No procedure for state/command-queue reconciliation after an outage	We could explore it
G3	No RCP-parameter treatment of the recovery interval	We could use the envelope for this
G4	No mapping between radio-layer re-establishment and operational link-loss procedure	Low-hanging fruit.
G5	Resilience formalism not applied to aviation communication links	We have to cite initial bibliography (Panteli, Sterbenz, Reed, Sharma) as the methods being transferred.
G6	No time-variant margin construct for VLL C2	We can link it to SORA's spatial contingency volume.
G7	No harmonized European C2 MASPS	The motivation rather than a gap. We use it to motivate the work; EUROCAE WG-105 SG-2 has acknowledged it.

Proactive—Though less developed than its reactive counterpart, there is prior work in anticipating the C2 link degradation. Yang *et al.* outline the telecom perspective on drone onnectivity [39]. For mobility management, researchers compare model-based and learning-based approaches [40], apply model-predictive control to predict handovers [41], and examine vertical handovers in non-terrestrial networks [42]. For prediction specifically, López *et al.* use measurements to estimate and predict UAV service outages [43], [44].

Post-corrective—Regulations dictate the response ot a LC2L: the aircraft must hold, return to a point, or terminate the flight. But they ignore the return path. There is no specification in the literature regarding a maximum time to re-establish the link. There is no criterion distinguishing a link that carries packets to resume normal control. There is no criteria to distinguish a barely functioning link from one trustworthy enough for normal control, and no procedure for reconciling vehicle state and command queue after an outage. Finally, there are no continuity or availability requirements specific to the recovery interval.

This gap is structural. 3GPP treats link re-establishment as a radio procedure, with its own detection and timers [45]. On the other hand, aviation treats link loss as an operational event, with contingency procedures. No document bridges the two. As a result, a C2 system inherits a radio recovery timeline it cannot control, alongside an operational obligation unknown to the radio layer. Similarly, SORA's taxonomy defines normal, abnormal, and emergency procedures, but omits the transition back to normal. This is the vital path for resynchronisation and state restoration.

3. METHODOLOGY

We adapt the RCP metrics to the resilience framework in [27] to construct the mathematical model. These are *Pre-disturbance (Φ)*, governed by link availability. *Degradation (Λ)*, the decision latency between RF degradation and the transition to LC2L state. Mapped to Continuity and Transaction Time. *Loss (E)* pending, most likely as loss of availability and *Recovery (Π)*, time to coordinate new clearance once the RF link is restored.

4. TODOS

- The rules of the paper are in Example_2027.pdf and core idea is in C2-Reliability-Resilience.pdf
- Main objective of the paper : (i) Resilience phases and methods (ii) C2 operational margin (iii) Validation with results

Task Breakdown - To dos

1. Background - The 3 phases idea comes from power engineering. Should go deep and do lit review on Methods
2. Do we have standards for post recovery phase procedure esp for communication in Aviation
3. what are the methods to try - mathematical framework
4. Parameters and KPIs for C2 Operational envelope
5. Testbed - evaluation requirements

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